

5.5 Another Inverse Problem for Boundary Value Problem

```
In [28]: using CairoMakie
using LinearAlgebra
using LinearSolve
using SparseArrays
using BenchmarkTools
using CSV
using DataFrames
```

In this notebook, you are given that some data is taken from a beam deflection

You are given some data points, which is the deflection, y_i at each location x_i , (x_i, y_i) . Your task is to find $w(x)$ that fits this data.

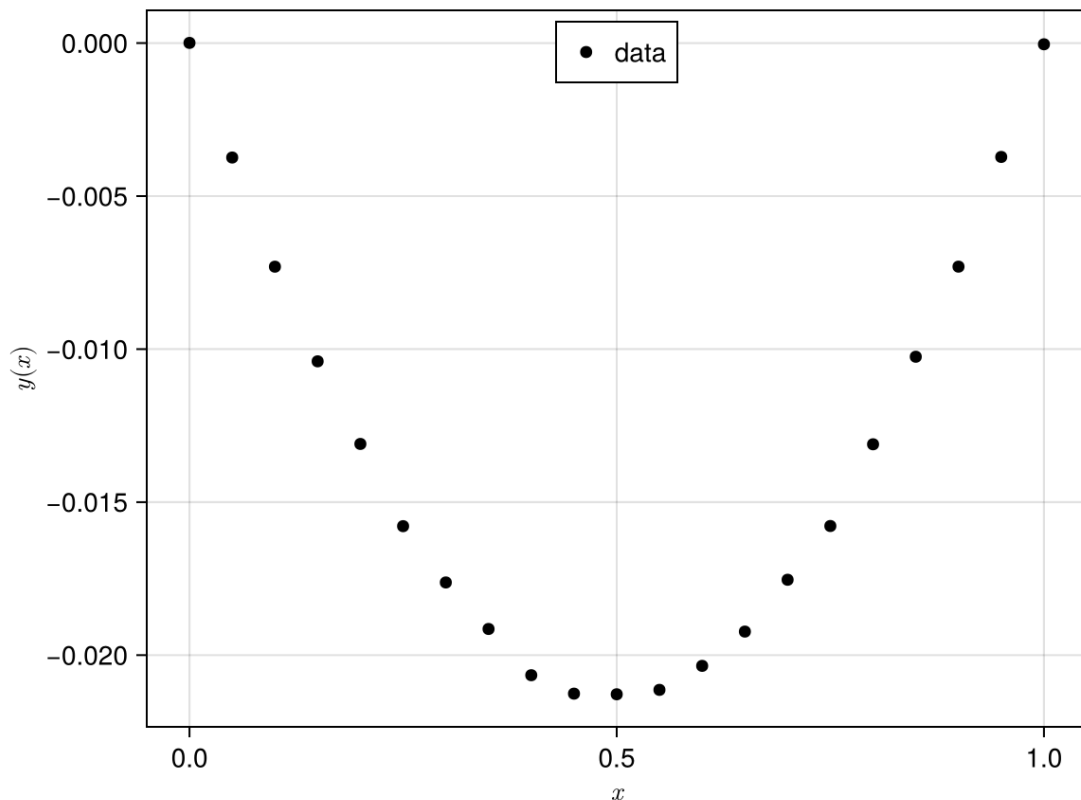
As always, let's ready the (x_i, y_i) data points and see how they look like.

```
In [29]: df = CSV.read("Data01.csv", DataFrame;
select=["xdata", "ydata"])
xdata=df[:, :xdata]
ydata=df[:, :ydata]
```

21-element Vector{Float64}:

```
 6.193274031415785e-6
-0.003739378905905285
-0.007306349221306566
-0.010399180465463597
-0.013098308162873173
-0.01578551634224239
-0.01762495953731014
-0.019144283606991372
-0.020655809722303967
-0.021258175295570867
 ⋮
-0.02035003021705797
-0.019230083080761153
-0.017536481088091162
-0.015780287203041334
-0.013109685125591448
-0.010249205411925406
-0.007305186584581027
-0.0037204446087818945
-3.915200830436959e-5
```

```
In [30]: fig = Figure()
ax = Axis(fig[1, 1], xlabel = L"x", ylabel = L"y(x)")
scatter!(ax, xdata, ydata, color=:black, label="data")
axislegend(ax, position=:ct)
save("../figures/05p05ParameterEstimationBVP0201.svg", fig)
fig
```



Since we know that our data is taken from the deflection of a beam, can use what we know about the deflection of the beam. From other parts of your degree program, we know that a good model for the beam deflection is

$$\frac{d^2 y}{dx^2} + 4y(x) = w(x)$$

y is the deflection of the beam and $w(x)$ a weight distribution on this beam. The domain is $x \in [0, 1]$. There is no deflection at the ends of the beam, so $y(0) = y(1) = 0.0$. We can discretise this equation as per usual

$$\frac{d^2 y}{dx^2} \approx \frac{y_{i-1} - 2y_i + y_{i+1}}{\Delta^2}$$

the governing equation for beam deflection becomes

$$\frac{y_{i-1} - 2y_i + y_{i+1}}{\Delta^2} + 4y_i = w(x_i)$$

We do not know what $w(x)$ is so we will just use a model and say

$$w(x) = a_0 + a_1 x$$

Our model becomes

$$\frac{y_{i-1} - 2y_i + y_{i+1}}{\Delta^2} + 4y_i = w(x_i) = a_0 + a_1 x_i$$

So our task now is to find a_0 and a_1 such that when you solve the the equation above, you get y_i that agrees best agree with your data.

For a domain of 5 intervals, our matrix equations will look like

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 1/\Delta^2 & -2/\Delta^2 + 4 & 1/\Delta^2 & 0 & 0 & 0 \\ 0 & 1/\Delta^2 & -2/\Delta^2 + 4 & 1/\Delta^2 & 0 & 0 \\ 0 & 0 & 1/\Delta^2 & -2/\Delta^2 + 4 & 1/\Delta^2 & 0 \\ 0 & 0 & 0 & 1/\Delta^2 & -2/\Delta^2 + 4 & 1/\Delta^2 \\ 0 & 0 & 0 & 0 & -1 & 1 \end{bmatrix} \begin{Bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \\ y_6 \end{Bmatrix} =$$

Everytime with want to make a prediction, we need to solve the equation above. With the exception of y_1 and y_6 , all your y_i 's above are dependent on a_0 and a_1 . The dependence is not explicit but there is an implicit relationship. In terms of a model, we can think of the above equations as an implicitly definition of our $f_{model}(a_0, a_1, x)$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 1/\Delta^2 & -2/\Delta^2 + 4 & 1/\Delta^2 & 0 & 0 & 0 \\ 0 & 1/\Delta^2 & -2/\Delta^2 + 4 & 1/\Delta^2 & 0 & 0 \\ 0 & 0 & 1/\Delta^2 & -2/\Delta^2 + 4 & 1/\Delta^2 & 0 \\ 0 & 0 & 0 & 1/\Delta^2 & -2/\Delta^2 + 4 & 1/\Delta^2 \\ 0 & 0 & 0 & 0 & -1 & 1 \end{bmatrix} \begin{Bmatrix} f_{model}(a_0, a_1, x_1) \\ f_{model}(a_0, a_1, x_2) \\ f_{model}(a_0, a_1, x_3) \\ f_{model}(a_0, a_1, x_4) \\ f_{model}(a_0, a_1, x_5) \\ f_{model}(a_0, a_1, x_6) \end{Bmatrix}$$

Set up matrices

```
In [31]: Delta=xdata[2]-xdata[1]
n=length(xdata)
d=zeros(n)
dl=zeros(n-1)
du=zeros(n-1)
C=zeros(n)
du[1]=0.0
d[1]=1.0
C[1]=0.0
for i=2:n-1
    dl[i-1]=(1.0/Delta^2)
    d[i]=-(2.0/Delta^2)+4.0
    du[i]=(1.0/Delta^2)
end
dl[n-1]=0.0
d[n]=1.0
C[n]=0.0

# Construct the specialized structured matrix
A = Tridiagonal(dl, d, du)
```


Current loss after 50 iterations: 0.4159506366936311
Current loss after 100 iterations: 0.37160558592396
Current loss after 150 iterations: 0.3309155298329799
Current loss after 200 iterations: 0.29368779452999194
Current loss after 250 iterations: 0.259729987851196
Current loss after 300 iterations: 0.2288530396570117
Current loss after 350 iterations: 0.20087131958566437
Current loss after 400 iterations: 0.17560279645712396
Current loss after 450 iterations: 0.1528692446764662
Current loss after 500 iterations: 0.13249649078900783
Current loss after 550 iterations: 0.11431469616493299
Current loss after 600 iterations: 0.09815867177414851
Current loss after 650 iterations: 0.08386822006508764
Current loss after 700 iterations: 0.07128849765892703
Current loss after 750 iterations: 0.06027039113943556
Current loss after 800 iterations: 0.050670896798267244
Current loss after 850 iterations: 0.042353493907833155
Current loss after 900 iterations: 0.03518850006014185
Current loss after 950 iterations: 0.029053396450528304
Current loss after 1000 iterations: 0.02383311081213822
Current loss after 1050 iterations: 0.019420246112682303
Current loss after 1100 iterations: 0.015715244169201042
Current loss after 1150 iterations: 0.01262647503453275
Current loss after 1200 iterations: 0.010070245322259632
Current loss after 1250 iterations: 0.007970721468579486
Current loss after 1300 iterations: 0.0062597671281219335
Current loss after 1350 iterations: 0.004876697269372476
Current loss after 1400 iterations: 0.003767954849197863
Current loss after 1450 iterations: 0.0028867189741705623
Current loss after 1500 iterations: 0.002192455986580525
Current loss after 1550 iterations: 0.0016504267741969745
Current loss after 1600 iterations: 0.001231164683589851
Current loss after 1650 iterations: 0.000909938675891837
Current loss after 1700 iterations: 0.0006662158318492129
Current loss after 1750 iterations: 0.00048313608498369223
Current loss after 1800 iterations: 0.00034701028282646274
Current loss after 1850 iterations: 0.00024685052247419017
Current loss after 1900 iterations: 0.00017393936429320797
Current loss after 1950 iterations: 0.0001214421734537731
Current loss after 2000 iterations: 8.406462547893271e-5
Current loss after 2050 iterations: 5.775545653594821e-5
Current loss after 2100 iterations: 3.94529194293368e-5
Current loss after 2150 iterations: 2.687216037380037e-5
Current loss after 2200 iterations: 1.8329862710328723e-5
Current loss after 2250 iterations: 1.2601986695817394e-5
Current loss after 2300 iterations: 8.810224322110682e-6
Current loss after 2350 iterations: 6.332828103706831e-6
Current loss after 2400 iterations: 4.735701940657206e-6
Current loss after 2450 iterations: 3.7200012452511047e-6
Current loss after 2500 iterations: 3.082925257807382e-6
Current loss after 2550 iterations: 2.6888522579058582e-6
Current loss after 2600 iterations: 2.4484334091010062e-6
Current loss after 2650 iterations: 2.3036982693425224e-6
Current loss after 2700 iterations: 2.217618517148223e-6
Current loss after 2750 iterations: 2.1669177195615654e-6
Current loss after 2800 iterations: 2.137201495678703e-6
Current loss after 2850 iterations: 2.11971603058626e-6
Current loss after 2900 iterations: 2.109228226441823e-6
Current loss after 2950 iterations: 2.1026640756149265e-6
Current loss after 3000 iterations: 2.0982499350619334e-6

Current loss after 3050 iterations: 2.0949809887661052e-6
Current loss after 3100 iterations: 2.092298452961328e-6
Current loss after 3150 iterations: 2.0898973326455765e-6
Current loss after 3200 iterations: 2.0876141890418526e-6
Current loss after 3250 iterations: 2.0853629407154563e-6
Current loss after 3300 iterations: 2.083098899665848e-6
Current loss after 3350 iterations: 2.0807990508889037e-6
Current loss after 3400 iterations: 2.078451473336492e-6
Current loss after 3450 iterations: 2.0760497909125853e-6
Current loss after 3500 iterations: 2.0735903281599575e-6
Current loss after 3550 iterations: 2.0710706863706613e-6
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Current loss after 4450 iterations: 2.013642046481964e-6
Current loss after 4500 iterations: 2.009679097663586e-6
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Current loss after 4600 iterations: 2.0014724107079195e-6
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Current loss after 4700 iterations: 1.992878532447623e-6
Current loss after 4750 iterations: 1.988431488148372e-6
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Current loss after 4850 iterations: 1.979227128244644e-6
Current loss after 4900 iterations: 1.9744657063011852e-6
Current loss after 4950 iterations: 1.9695953409722293e-6
Current loss after 5000 iterations: 1.964613913120067e-6
Current loss after 5050 iterations: 1.9595192782380047e-6
Current loss after 5100 iterations: 1.954309267114796e-6
Current loss after 5150 iterations: 1.948981686570226e-6
Current loss after 5200 iterations: 1.943534320261946e-6
Current loss after 5250 iterations: 1.937964929561618e-6
Current loss after 5300 iterations: 1.932271254522951e-6
Current loss after 5350 iterations: 1.9264510149190774e-6
Current loss after 5400 iterations: 1.920501911377711e-6
Current loss after 5450 iterations: 1.9144216265984145e-6
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Current loss after 5650 iterations: 1.8887417781995658e-6
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Current loss after 5750 iterations: 1.8750534374064035e-6
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Current loss after 5850 iterations: 1.8607739694375175e-6
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Current loss after 6200 iterations: 1.805863198667408e-6
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Current loss after 6450 iterations: 1.7616112863269525e-6
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Current loss after 6550 iterations: 1.7426618079564907e-6
Current loss after 6600 iterations: 1.7329107637648177e-6
Current loss after 6650 iterations: 1.7229727422863116e-6
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Current loss after 6750 iterations: 1.7025277765088948e-6
Current loss after 6800 iterations: 1.6920169927400204e-6
Current loss after 6850 iterations: 1.6813115636607442e-6
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Current loss after 6950 iterations: 1.6593098972865906e-6
Current loss after 7000 iterations: 1.648010437366483e-6
Current loss after 7050 iterations: 1.636509899477586e-6
Current loss after 7100 iterations: 1.6248069145307277e-6
Current loss after 7150 iterations: 1.6129002223931637e-6
Current loss after 7200 iterations: 1.600788678924189e-6
Current loss after 7250 iterations: 1.5884712632022255e-6
Current loss after 7300 iterations: 1.5759470849613366e-6
Current loss after 7350 iterations: 1.5632153922133757e-6
Current loss after 7400 iterations: 1.5502755790650661e-6
Current loss after 7450 iterations: 1.5371271937132713e-6
Current loss after 7500 iterations: 1.523769946612464e-6
Current loss after 7550 iterations: 1.5102037188026894e-6
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Current loss after 7650 iterations: 1.4824447491984725e-6
Current loss after 7700 iterations: 1.4682526994439459e-6
Current loss after 7750 iterations: 1.4538530706607092e-6
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Current loss after 7900 iterations: 1.4094184730705894e-6
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Current loss after 8050 iterations: 1.3631606448161483e-6
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Current loss after 8250 iterations: 1.2987476575591166e-6
Current loss after 8300 iterations: 1.2821752672733296e-6
Current loss after 8350 iterations: 1.2654223837230597e-6
Current loss after 8400 iterations: 1.248493139812402e-6
Current loss after 8450 iterations: 1.2313919956386707e-6
Current loss after 8500 iterations: 1.2141237457247724e-6
Current loss after 8550 iterations: 1.1966935258249964e-6
Current loss after 8600 iterations: 1.1791068192551932e-6
Current loss after 8650 iterations: 1.161369462684522e-6
Current loss after 8700 iterations: 1.1434876513361e-6
Current loss after 8750 iterations: 1.1254679435381225e-6
Current loss after 8800 iterations: 1.107317264557081e-6
Current loss after 8850 iterations: 1.0890429096616413e-6
Current loss after 8900 iterations: 1.070652546338806e-6
Current loss after 8950 iterations: 1.052154215607082e-6
Current loss after 9000 iterations: 1.0335563323560636e-6

Current loss after 9050 iterations: 1.014867684641892e-6
Current loss after 9100 iterations: 9.960974318760565e-7
Current loss after 9150 iterations: 9.7725510183706e-7
Current loss after 9200 iterations: 9.583505864346163e-7
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Current loss after 10250 iterations: 5.678142271153051e-7
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Current loss after 10350 iterations: 5.336765173398156e-7
Current loss after 10400 iterations: 5.16970586267577e-7
Current loss after 10450 iterations: 5.005240331126834e-7
Current loss after 10500 iterations: 4.843495595406575e-7
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Current loss after 10600 iterations: 4.5286587076196136e-7
Current loss after 10650 iterations: 4.375801392298313e-7
Current loss after 10700 iterations: 4.22613366478354e-7
Current loss after 10750 iterations: 4.079760705788916e-7
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Current loss after 10900 iterations: 3.6613747182015363e-7
Current loss after 10950 iterations: 3.529113093684976e-7
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Current loss after 11050 iterations: 3.2758377037192813e-7
Current loss after 11100 iterations: 3.15494654169687e-7
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Current loss after 12750 iterations: 1.2059036762939234e-7
Current loss after 12800 iterations: 1.1927676343964015e-7
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Current loss after 14350 iterations: 1.1019564509374403e-7
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Current loss after 14500 iterations: 1.1018687522688916e-7
Current loss after 14550 iterations: 1.1018529574960169e-7
Current loss after 14600 iterations: 1.1018413511270107e-7
Current loss after 14650 iterations: 1.1018328965522995e-7
Current loss after 14700 iterations: 1.1018267928004094e-7
Current loss after 14750 iterations: 1.1018224265762364e-7
Current loss after 14800 iterations: 1.1018193326468788e-7
Current loss after 14850 iterations: 1.1018171614552729e-7
Current loss after 14900 iterations: 1.1018156529257964e-7
Current loss after 14950 iterations: 1.1018146154952352e-7
Current loss after 15000 iterations: 1.101813909517026e-7

Current loss after 15050 iterations: 1.1018134342635389e-7
Current loss after 15100 iterations: 1.1018131178675028e-7
Current loss after 15150 iterations: 1.1018129096227119e-7
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Current loss after 15300 iterations: 1.1018126318354845e-7
Current loss after 15350 iterations: 1.1018125972001261e-7
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Current loss after 15500 iterations: 1.1018125548311783e-7
Current loss after 15550 iterations: 1.1018125501614798e-7
Current loss after 15600 iterations: 1.1018125474309678e-7
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Current loss after 35700 iterations: 1.10181254388717e-7
Current loss after 35750 iterations: 1.101812543886961e-7
Current loss after 35800 iterations: 1.1018125438871472e-7
Current loss after 35850 iterations: 1.1018125438872847e-7
Current loss after 35900 iterations: 1.1021236202740783e-7
Current loss after 35950 iterations: 1.1030828801492325e-7
Current loss after 36000 iterations: 1.1018127294251355e-7

Current loss after 36050 iterations: 1.1018125924699744e-7
Current loss after 36100 iterations: 1.101812544639057e-7
Current loss after 36150 iterations: 1.1018125438874115e-7
Current loss after 36200 iterations: 1.1018125438870707e-7
Current loss after 36250 iterations: 1.1018125438869793e-7
Current loss after 36300 iterations: 1.1018125438869518e-7
Current loss after 36350 iterations: 1.1018125438870662e-7
Current loss after 36400 iterations: 1.1018130442589205e-7
Current loss after 36450 iterations: 1.1024718878047522e-7
Current loss after 36500 iterations: 1.1018735270932748e-7
Current loss after 36550 iterations: 1.1018127722353999e-7
Current loss after 36600 iterations: 1.101812544061726e-7
Current loss after 36650 iterations: 1.101812543887657e-7
Current loss after 36700 iterations: 1.1018125438871276e-7
Current loss after 36750 iterations: 1.1018125438868234e-7
Current loss after 36800 iterations: 1.1018125438872115e-7
Current loss after 36850 iterations: 1.1018125438871609e-7
Current loss after 36900 iterations: 1.1018125438870327e-7
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Current loss after 37050 iterations: 1.1254025300812497e-7
Current loss after 37100 iterations: 1.1019502231244012e-7
Current loss after 37150 iterations: 1.1018127544762274e-7
Current loss after 37200 iterations: 1.1018125467427221e-7
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Current loss after 37300 iterations: 1.1018125438872016e-7
Current loss after 37350 iterations: 1.10181254388705e-7
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Current loss after 37450 iterations: 1.1018125438868973e-7
Current loss after 37500 iterations: 1.1018125438869869e-7
Current loss after 37550 iterations: 1.1018125438870364e-7
Current loss after 37600 iterations: 1.101812543887014e-7
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Current loss after 37750 iterations: 1.1018135033164554e-7
Current loss after 37800 iterations: 1.1018125656032928e-7
Current loss after 37850 iterations: 1.1018125440451743e-7
Current loss after 37900 iterations: 1.1018125438880008e-7
Current loss after 37950 iterations: 1.1018125438871075e-7
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Current loss after 38050 iterations: 1.1018125438870462e-7
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Current loss after 38150 iterations: 1.1018125542506481e-7
Current loss after 38200 iterations: 1.1216056013451558e-7
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Current loss after 39000 iterations: 1.1018125438871982e-7

Current loss after 39050 iterations: 1.101812543887002e-7
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Current loss after 39950 iterations: 1.101812544177616e-7
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Current loss after 40050 iterations: 1.1018125438868363e-7
Current loss after 40100 iterations: 1.1018125438868879e-7
Current loss after 40150 iterations: 1.1018125438870948e-7
Current loss after 40200 iterations: 1.1018125438870488e-7
Current loss after 40250 iterations: 1.1018125467306481e-7
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Current loss after 40450 iterations: 1.1018125479959061e-7
Current loss after 40500 iterations: 1.101812543894801e-7
Current loss after 40550 iterations: 1.1018125438871712e-7
Current loss after 40600 iterations: 1.1018125438869889e-7
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Current loss after 40700 iterations: 1.1018125438872234e-7
Current loss after 40750 iterations: 1.1018125472744224e-7
Current loss after 40800 iterations: 1.1241226196420903e-7
Current loss after 40850 iterations: 1.1019070811303485e-7
Current loss after 40900 iterations: 1.1018126151685622e-7
Current loss after 40950 iterations: 1.1018125441473548e-7
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Current loss after 41050 iterations: 1.1018125438869784e-7
Current loss after 41100 iterations: 1.1018125438871832e-7
Current loss after 41150 iterations: 1.1018125438869684e-7
Current loss after 41200 iterations: 1.1018125438867615e-7
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Current loss after 41350 iterations: 1.1018146363097287e-7
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Current loss after 41550 iterations: 1.1018125438871484e-7
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Current loss after 41650 iterations: 1.1018125438869386e-7
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Current loss after 41950 iterations: 1.1018125440019866e-7
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Current loss after 43550 iterations: 1.101812543887151e-7
Current loss after 43600 iterations: 1.1018125438869661e-7
Current loss after 43650 iterations: 1.1018125443240565e-7
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Current loss after 43750 iterations: 1.1019572794817696e-7
Current loss after 43800 iterations: 1.1018133840765289e-7
Current loss after 43850 iterations: 1.1018125477382686e-7
Current loss after 43900 iterations: 1.1018125438878976e-7
Current loss after 43950 iterations: 1.1018125438872893e-7
Current loss after 44000 iterations: 1.1018125438871781e-7
Current loss after 44050 iterations: 1.1018125438867585e-7
Current loss after 44100 iterations: 1.101812543886988e-7
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Current loss after 44250 iterations: 1.1018258786782652e-7
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Current loss after 44400 iterations: 1.1018125438873723e-7
Current loss after 44450 iterations: 1.1018125438873077e-7
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Current loss after 44550 iterations: 1.101812543886985e-7
Current loss after 44600 iterations: 1.1018128838722659e-7
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Current loss after 45000 iterations: 1.1018125438870741e-7

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Current loss after 45700 iterations: 1.1018125498629733e-7
Current loss after 45750 iterations: 1.1018125439314893e-7
Current loss after 45800 iterations: 1.101812543887554e-7
Current loss after 45850 iterations: 1.1018125438871252e-7
Current loss after 45900 iterations: 1.1018125438870655e-7
Current loss after 45950 iterations: 1.1018125438870324e-7
Current loss after 46000 iterations: 1.4092681047984755e-7
Current loss after 46050 iterations: 1.1019143897517578e-7
Current loss after 46100 iterations: 1.1018171889472918e-7
Current loss after 46150 iterations: 1.1018125775633837e-7
Current loss after 46200 iterations: 1.1018125439187087e-7
Current loss after 46250 iterations: 1.1018125438876697e-7
Current loss after 46300 iterations: 1.1018125438868844e-7
Current loss after 46350 iterations: 1.1018125438869366e-7
Current loss after 46400 iterations: 1.1018125438869215e-7
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Current loss after 47000 iterations: 1.109251850672395e-7
Current loss after 47050 iterations: 1.101819076878619e-7
Current loss after 47100 iterations: 1.1018126024945578e-7
Current loss after 47150 iterations: 1.1018125452417465e-7
Current loss after 47200 iterations: 1.101812543888884e-7
Current loss after 47250 iterations: 1.1018125438869968e-7
Current loss after 47300 iterations: 1.1018125438869932e-7
Current loss after 47350 iterations: 1.1018125438869629e-7
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Current loss after 47450 iterations: 1.1035991940707993e-7
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Current loss after 47600 iterations: 1.1018125447352013e-7
Current loss after 47650 iterations: 1.1018125438908466e-7
Current loss after 47700 iterations: 1.1018125438873661e-7
Current loss after 47750 iterations: 1.1018125438871019e-7
Current loss after 47800 iterations: 1.1018125438871059e-7
Current loss after 47850 iterations: 1.101812543886838e-7
Current loss after 47900 iterations: 1.1018321706560173e-7
Current loss after 47950 iterations: 1.1032503318797742e-7
Current loss after 48000 iterations: 1.1018394138553873e-7

```

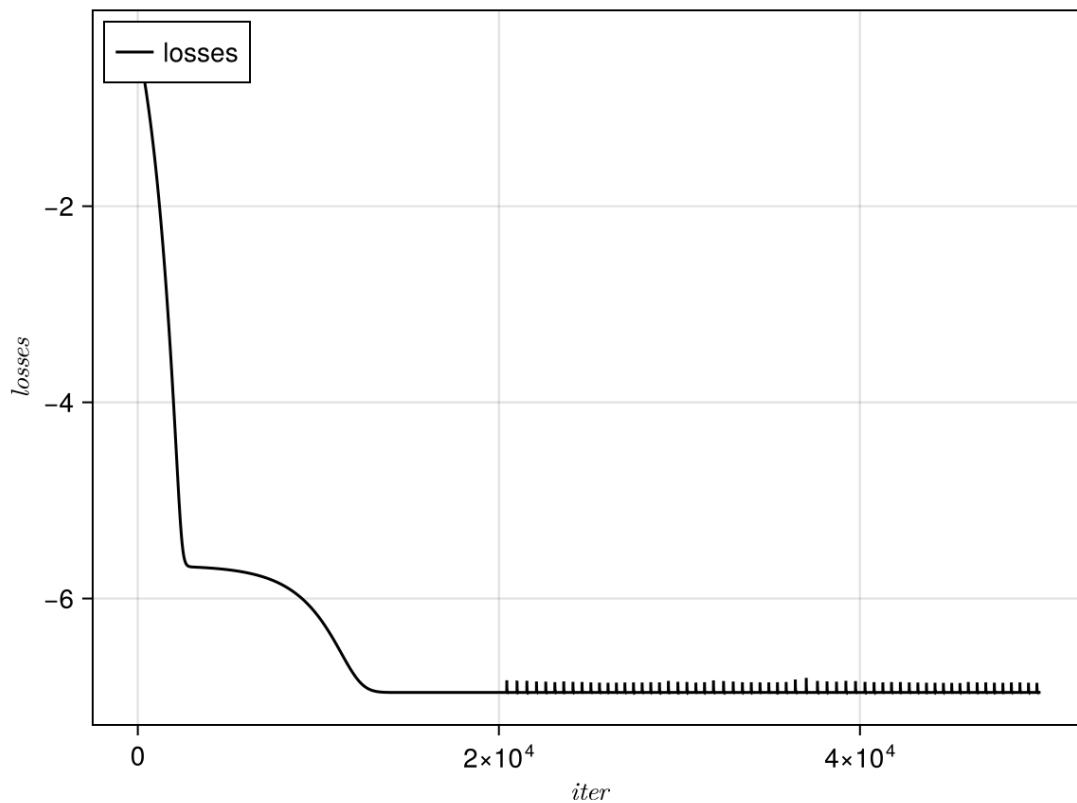
Current loss after 48050 iterations: 1.1018125752749131e-7
Current loss after 48100 iterations: 1.1018125445691953e-7
Current loss after 48150 iterations: 1.1018125438935812e-7
Current loss after 48200 iterations: 1.1018125438870366e-7
Current loss after 48250 iterations: 1.1018125438869018e-7
Current loss after 48300 iterations: 1.1018125438870971e-7
Current loss after 48350 iterations: 1.1018125440487576e-7
Current loss after 48400 iterations: 1.102268738399046e-7
Current loss after 48450 iterations: 1.1018359845492234e-7
Current loss after 48500 iterations: 1.1018125722695332e-7
Current loss after 48550 iterations: 1.1018125440257603e-7
Current loss after 48600 iterations: 1.1018125439051424e-7
Current loss after 48650 iterations: 1.1018125438870449e-7
Current loss after 48700 iterations: 1.1018125438868773e-7
Current loss after 48750 iterations: 1.1018125438870337e-7
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Current loss after 48850 iterations: 1.1025189862727288e-7
Current loss after 48900 iterations: 1.1042163169186757e-7
Current loss after 48950 iterations: 1.101817159981427e-7
Current loss after 49000 iterations: 1.1018126446052263e-7
Current loss after 49050 iterations: 1.1018125440665464e-7
Current loss after 49100 iterations: 1.101812543887047e-7
Current loss after 49150 iterations: 1.1018125438870523e-7
Current loss after 49200 iterations: 1.1018125438870343e-7
Current loss after 49250 iterations: 1.1018125438869935e-7
Current loss after 49300 iterations: 1.10181254414783e-7
Current loss after 49350 iterations: 1.1056267220320327e-7
Current loss after 49400 iterations: 1.1018810599873017e-7
Current loss after 49450 iterations: 1.1018133950041172e-7
Current loss after 49500 iterations: 1.1018125514162489e-7
Current loss after 49550 iterations: 1.1018125438944348e-7
Current loss after 49600 iterations: 1.1018125438873134e-7
Current loss after 49650 iterations: 1.1018125438870169e-7
Current loss after 49700 iterations: 1.101812543886902e-7
Current loss after 49750 iterations: 1.1018125438870053e-7
Current loss after 49800 iterations: 1.3398476999067073e-7
Current loss after 49850 iterations: 1.1020018634053055e-7
Current loss after 49900 iterations: 1.1018127055705172e-7
Current loss after 49950 iterations: 1.101812543933311e-7
Current loss after 50000 iterations: 1.1018125439264511e-7
retcode: Default
u: 2-element Vector{Float64}:
 0.10202966301193382
-0.0037530782220365428

```

```

In [35]: fig = Figure()
ax = Axis(fig[1, 1], xlabel = L"iter", ylabel = L"losses")
lines!(ax, log10.(losses), color=:black, label="losses")
axislegend(ax, position=:lt)
fig

```



In [36]: `sol.u`

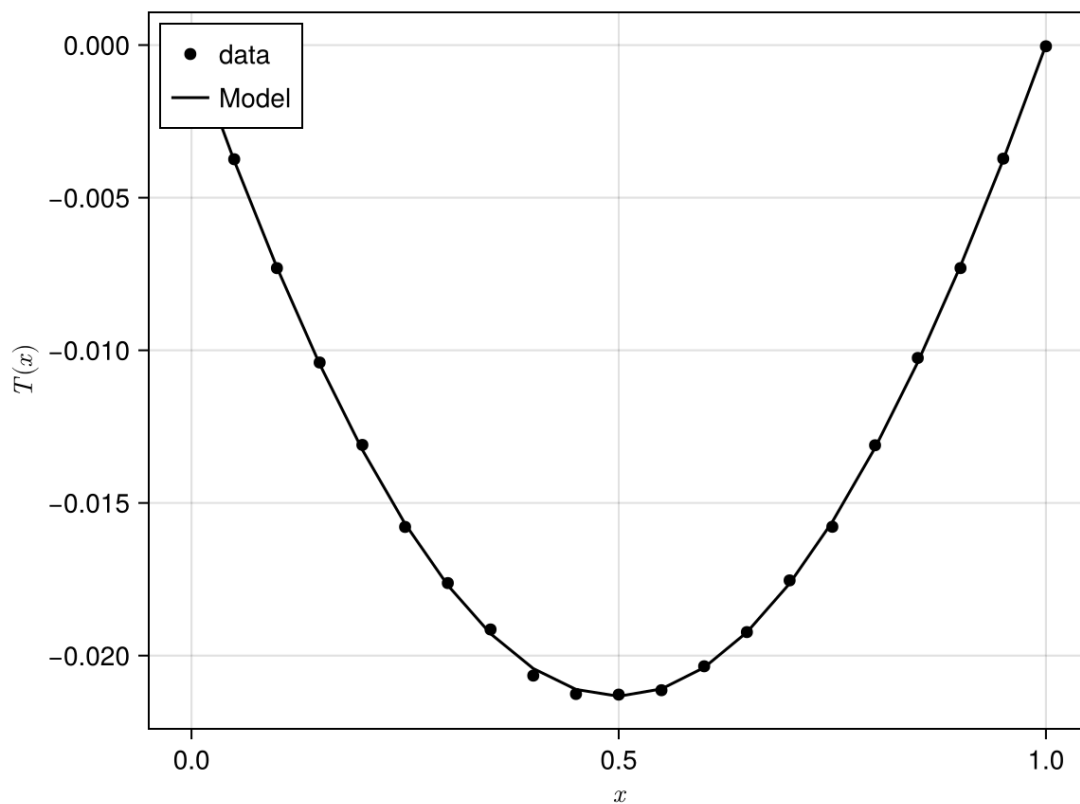
```
2-element Vector{Float64}:
 0.10202966301193382
-0.0037530782220365428
```

In [37]: `C=(sol.u[1].+sol.u[2]*xdata)`
`C[1]=0.0`
`C[n]=0.0`
`prob=LinearProblem(A,C)`
`ymodel=LinearSolve.solve(prob)`

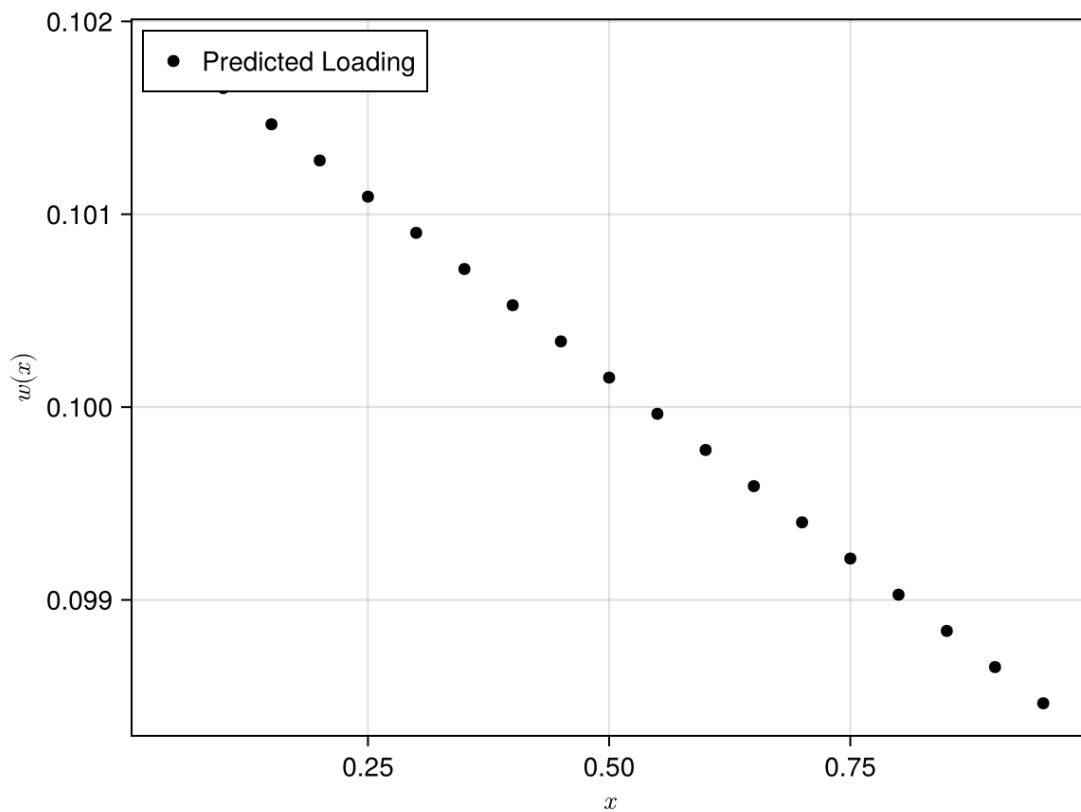
retcode: Success

```
u: 21-element Vector{Float64}:
 3.6748382115092687e-16
-0.0037874982794743
-0.007282516553402144
-0.01045057377382164
-0.013260458503306347
-0.015684541029339173
-0.017699049661437545
-0.019284308448058233
-0.020424933936112784
-0.021109989005498406
⋮
-0.020391032753185618
-0.019236228895698294
-0.01764008734383496
-0.015619038647892
-0.013193762429606603
-0.010388980585939374
-0.007233210070104745
-0.0037584777220388155
 0.0
```

```
In [38]: fig = Figure()
ax = Axis(fig[1, 1], xlabel = L"x", ylabel = L"T(x)")
scatter!(ax,xdata,ydata,color=:black,label="data")
lines!(ax,xdata,ymodel,color=:black,label="Model")
axislegend(ax,position=:lt)
fig
```



```
In [39]: fig = Figure()
ax = Axis(fig[1, 1], xlabel = L"x", ylabel = L"w(x)")
scatter!(ax,xdata[2:end-1],C[2:end-1],color=:black,label="Predicted Loadi")
axislegend(ax,position=:lt)
fig
```



In [40]: `sol.u`

```
2-element Vector{Float64}:
 0.10202966301193382
-0.0037530782220365428
```

Class Exercise: Now try to model the data using simple polynomial.

$$f_{\text{model}}(\vec{a}, x) = a_0x + a_1x + a_2x^2 + a_3x^3 + \dots$$

What are the values of $\vec{a}$ that best fit this data?

In []: